

09b - Deployment Fixture Pack and Mutation Matrix v0.1

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Date	2026-07-05
Author	Kotov Ivan, Bruxelles, Belgique - 2026
ORCID	0009-0009-6002-9845
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Abstract / purpose

09 defines the deployment and regulated-release boundary. 09a defines the first machine-facing deployment packet and checker-seed contract. 09b hardens that contract with an adversarial fixture pack and mutation matrix. The governing question is narrow: The answer must be executable enough for future checker work: 09b does not add a new deployment right. It pressures the 09a boundary. ---

Non-claims boundary

- _ not legal advice
- _ not privacy-law certification
- _ not safety certification
- _ not deployment authorization
- _ not standards compliance certification
- _ not regulated approval
- _ not C-A1 ratification
- _ not personhood proof
- _ not consciousness proof
- _ not live substrate truth
- _ not proof of completeness

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09b - Deployment Fixture Pack and Mutation Matrix v0.1

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0. Purpose

09 defines the deployment and regulated-release boundary.

09a defines the first machine-facing deployment packet and checker-seed contract.

09b hardens that contract with an adversarial fixture pack and mutation matrix.

The governing question is narrow:

Can the 09a deployment gate reject the common ways deployment authority is laundered from publication, fixtures, reproduction, external review, standards mapping, string-only approver roles, or bureaucratic paperwork?

The answer must be executable enough for future checker work:

fixture case -> expected decision + finding codes
mutation -> disable one guard -> targeted fixture must catch it
coverage -> every guard has at least one fixture and one mutation probe

09b does not add a new deployment right. It pressures the 09a boundary.

0.1 Core rule

No deployment gate is credible until adversarial fixtures try to launder authority through it.

Operational expansion:

publication != deployment
checker pass != safety certification
fixture pass != production readiness
reproduction pass != certification
standards mapping != compliance
regulated submission != approval
institutional interest != endorsement
string role != attested approver
model/tool review != owner approval
negative-cache hit != allowed retry
red-pattern hit != allowed release
paperwork != +

0.2 Relation to +

09b includes explicit fixtures for the most important deployment-layer mistake:

deployment paperwork -> computes / replaces / ratifies +

Expected result:

PLUS_REPLACED_BY_DEPLOYMENT_RECORD -> DENY_RELEASE

The reason is structural. + is the governed non-collapsing binding boundary in $c = a + b$. A deployment record may govern release surfaces, but it cannot create the anchor-substrate binding, cannot replace the anchor, and cannot reduce the + boundary to a bureaucratic approval flag.

Compact rule:

09b may test deployment records.
09b may not turn deployment records into +.

1. Source bindings

09b is downstream of the active deployment stack.

```
| Binding | Artifact | SHA-256 | Role |
|---|---|---|---|
| `DOC04_CONTINUITY_STACK_UMBRELLA` | `CCALC_DOC04_CONTINUITY_STACK_UMBRELLA_v0_1.zip` |
| `6caeab8d489aaa6b26902db3bd3e4c5169e5c8d5617b93aed6586ea900322d` | continuity / equivalence / trace classification |
| `DOC05_SELF_EVO_STACK_UMBRELLA` | `CCALC_DOC05_SELF_EVO_STACK_UMBRELLA_v0_1.zip` |
| `9a0717809029f49df9001dc8ffa1803d9e9bfa1a824d317eebb146cbc7df43b4` | bounded growth / promotion / watch |
| `DOC06_RUNTIME_AUTHORITY_STACK_UMBRELLA` | `CCALC_DOC06_RUNTIME_AUTHORITY_STACK_UMBRELLA_v0_1.zip` |
| `ab95633f1b90f9d032fd231d6cbef3e71b7fe106e0a7fb61d198c2254fc7d307` | runtime authority / session / revocation |
| `DOC07_PUBLIC_EVIDENCE_STACK_UMBRELLA` | `CCALC_DOC07_PUBLIC_EVIDENCE_STACK_UMBRELLA_v0_1.zip` |
| `b25646d95e45f8a36e5610208d23a535d4c340484431d05efd7f1bf2389fdea3` | public evidence / custody / correction / citation sync |
| `DOC08_INTEROPERABILITY_STACK_UMBRELLA` | `CCALC_DOC08_INTEROPERABILITY_STACK_UMBRELLA_v0_1.zip` |
| `6015526f0ed49519e00c697a5ed375d37fe1aadf222c095f0facf79cb11e669f` | external review / patch / reproduction / disclosure |
| `DOC09_DEPLOYMENT_BOUNDARY_MD` | `09_C_DEPLOYMENT_PROFILE_AND_REGULATED_RELEASE_BOUNDARY_v0_1.md` |
| `16c888da5c4281c24f848246716b6f9f37d15236b909d66fc509090b5e7fd86d` | normative deployment boundary |
| `DOC09A_DEPLOYMENT_PROFILE_SCHEMA_MD` | `09a_DEPLOYMENT_PROFILE_SCHEMA_AND_CHECKER_SEED_v0_1.md` |
| `c169c6d46fb3f870e3ab8b8e18d29a7797849ee012923ba258d4a537cee7f98e` | machine-facing deployment packet/checker seed |
```

A missing or stale source binding invalidates release use of this package.

2. Non-claims

09b does not claim:

```
legal advice
privacy-law certification
safety certification
deployment authorization
regulated approval
standards compliance certification
C-A1 ratification
personhood or consciousness proof
live substrate truth
proof of completeness
operator identity or continuity
that fixtures exhaust adversarial behavior
that mutation coverage proves full correctness
that deployment paperwork computes +
```

It is a fixture/mutation hardening layer for the 09a deployment gate.

3. Package contents

This package contains:

```
09b_DEPLOYMENT_FIXTURE_PACK_AND_MUTATION_MATRIX_v0_1.md
SOURCE_BINDINGS.tsv
FIXTURE_CATALOG.tsv
MUTATION_MATRIX.tsv
COVERAGE_MATRIX.tsv
registry/GUARD_REGISTRY.tsv
fixtures/cases/*.json
src/deployment_fixture_mutation_checker_v0_1.py
scripts/run_09b_fixtures.py
scripts/run_09b_mutations.py
SHA256SUMS.txt
```

The checker is stdlib-only and closed-box. It validates the fixture catalog and mutation matrix. It is not a full deployment checker.

4. Record surfaces exercised

09b exercises the first 09a packet family:

```
DeploymentCandidatePacket
IntendedUserProfile
DeploymentModeRecord
ReleaseClassRecord
RegulatedSurfaceClassificationRecord
DeploymentAuthoritySurfaceMap
EvidenceCoverageRecord
ApproverAttestationRecord
DeploymentDecisionRecord
NextReleaseAdmissionRecord
```

It does not replace the schemas. It provides adversarial examples that later schema/checker implementations must reject or hold.

5. Guard registry

The normative seed guard set is stored in registry/GUARD_REGISTRY.tsv.

Guard	Finding code	Default decision	Meaning
---	---	---	---
'B09b-01'	'SOURCE_BINDING_MISSING'	'HOLD_FOR_EVIDENCE'	source bindings are mandatory
'B09b-02'	'INTENDED_USE_MISSING'	'HOLD_FOR_EVIDENCE'	intended use is mandatory
'B09b-03'	'MODE_RELEASE_CLASS_MISMATCH'	'HOLD_FOR_EVIDENCE'	deployment mode and release class must not contradict
'B09b-04'	'REGULATED_SURFACE_UNKNOWN'	'HOLD_FOR_EVIDENCE'	unknown regulated surface cannot be treated as low risk
'B09b-05'	'AUTHORITY_SURFACE_UNKNOWN'	'HOLD_FOR_EVIDENCE'	unknown authority surface cannot be allowed
'B09b-06'	'APPROVER_ATTESTATION_MISSING'	'REQUIRE_ANCHOR_ATTESTATION'	human/owner/regulatory approver roles require non-string attestation
'B09b-07'	'MODEL_OR_TOOL_APPROVER_INVALID'	'DENY_RELEASE'	model/tool approver cannot authorize deployment
'B09b-08'	'C_A1_OR_SAFETY_OVERCLAIM'	'DENY_RELEASE'	C-A1, safety, legal, deployment, or regulatory overclaims are denied
'B09b-09'	'RUNTIME_AUTHORITY_EVIDENCE_MISSING'	'HOLD_FOR_EVIDENCE'	live/runtime deployment requires 06 authority evidence
'B09b-10'	'SELF_EVO_WATCH_MISSING'	'HOLD_FOR_EVIDENCE'	growth-origin deployment requires 05/05d watch evidence
'B09b-11'	'CONTINUITY_UNKNOWN_OR_RUPTURE'	'HOLD_FOR_EVIDENCE'	continuity-dependent deployment cannot proceed on UNKNOWN/RUPTURE
'B09b-12'	'PUBLIC_CITATION_OR_RELEASE_DRIFT'	'HOLD_FOR_EVIDENCE'	public release/citation drift blocks dependency use
'B09b-13'	'INTEROP_CONFLICT_OPEN'	'HOLD_FOR_EVIDENCE'	open 08 high/critical conflict blocks deployment use
'B09b-14'	'CHECKER_OR_FIXTURE_PASS_AS_SAFETY'	'DENY_RELEASE'	fixture/checker pass is not safety certification
'B09b-15'	'REPRODUCTION_PASS_AS_CERTIFICATION'	'DENY_RELEASE'	reproduction pass is not certification/deployment authority
'B09b-16'	'STANDARDS_MAPPING_AS_COMPLIANCE'	'DENY_RELEASE'	standards mapping is not compliance/certification
'B09b-17'	'REGULATED_SUBMISSION_AS_APPROVAL'	'DENY_RELEASE'	regulated submission is not regulated approval
'B09b-18'	'PUBLICATION_AS_DEPLOYMENT'	'DENY_RELEASE'	DOI/publication/public release is not deployment authority
'B09b-19'	'ROLLBACK_MISSING_FOR_PRODUCTION'	'HOLD_FOR_EVIDENCE'	production/field release requires rollback/withdrawal route
'B09b-20'	'WATCH_MISSING_FOR_FIELD_OR_PRODUCTION'	'HOLD_FOR_EVIDENCE'	field/production/regulated release requires watch window
'B09b-21'	'NEGATIVE_CACHE_HIT_ALLOWED'	'QUARANTINE_RELEASE'	negative-cache hit blocks allow paths
'B09b-22'	'RED_PATTERN_ALLOWED'	'QUARANTINE_RELEASE'	unresolved red pattern blocks allow paths
'B09b-23'	'PLUS_REPLACED_BY_DEPLOYMENT_RECORD'	'DENY_RELEASE'	deployment paperwork may not compute or replace +
'B09b-24'	'INSTITUTIONAL_INTEREST_AS_ENDORSEMENT'	'HOLD_FOR_EVIDENCE'	institutional interest is not endorsement/approval
'B09b-25'	'NEXT_RELEASE_DIRTY_CARRYOVER'	'HOLD_FOR_EVIDENCE'	dirty close/failed watch must carry into next-release gate

Critical guard failures block allow decisions. A fixture may choose a stricter expected decision than the default. It may not choose a more permissive decision.

6. Fixture classes

The fixture catalog contains positive, negative, and negative-control fixtures.

6.1 Positive fixtures

Positive fixtures show what may pass within bounded scope:

```
valid_corpus_only_release
valid_checker_seed_release
valid_internal_trial_offline
valid_owner_field_pilot_with_anchor
valid_limited_production_with_watch
valid_regulated_submission_only
```

Important distinction:

```
valid_regulated_submission_only -> REQUIRE_REGULATED_REVIEW
```

A regulated submission packet may be admissible as a submission. It is not approval.

6.2 Negative-control fixture

```
allow_c_a10_control_token
```

This fixture proves that C-A10 is not rejected by naive prefix logic as C-A1.

6.3 Negative fixtures

Negative fixtures pressure the high-risk deployment laundering paths:

```
hold_missing_source_binding
hold_missing_intended_use
hold_mode_release_class_mismatch
hold_unknown_regulated_surface
hold_unknown_authority_surface
deny_string_only_anchor_role
deny_model_approver
deny_tool_approver
deny_c_a1_overclaim
deny_safety_certification_overclaim
hold_runtime_authority_missing
hold_post_promotion_watch_missing
hold_continuity_unknown
deny_continuity_rupture
hold_public_citation_drift
hold_retracted_public_artifact_dependency
hold_interop_conflict_open
deny_checker_pass_as_safety
deny_fixture_pass_as_production_ready
deny_reproduction_pass_as_certification
deny_standards_mapping_as_compliance
```

```

deny_regulated_submission_as_approval
deny_publication_as_deployment
deny_doi_as_deployment
hold_rollback_missing_for_production
hold_watch_missing_for_field_deployment
quarantine_negative_cache_hit_allowed
quarantine_red_pattern_allowed
deny_plus_replaced_by_deployment_record
deny_deployment_approval_as_binding_plus
hold_institutional_interest_as_non_authority
hold_dirty_next_release_carryover
deny_legal_certification_overclaim

```

7. Mutation matrix

The mutation matrix disables one guard at a time and reruns targeted fixtures.

A mutation is caught only when disabling the guard changes a targeted invalid fixture's result.

Mutation	Disabled guard	Target cases
MUT_ALLOW_MISSING_SOURCE_BINDING`	`B09b-01`	`hold_missing_source_binding`
MUT_ALLOW_MISSING_INTENDED_USE`	`B09b-02`	`hold_missing_intended_use`
MUT_ALLOW_MODE_RELEASE_MISMATCH`	`B09b-03`	`hold_mode_release_class_mismatch`
MUT_ALLOW_UNKNOWN_REGULATED_SURFACE`	`B09b-04`	`hold_unknown_regulated_surface`
MUT_ALLOW_UNKNOWN_AUTHORITY_SURFACE`	`B09b-05`	`hold_unknown_authority_surface`
MUT_ALLOW_STRING_ONLY_APPROVER`	`B09b-06`	`deny_string_only_anchor_role`
MUT_ALLOW_MODEL_TOOL_APPROVER`	`B09b-07`	`deny_model_approver, deny_tool_approver`
MUT_ALLOW_C_A1_CLAIM`	`B09b-08`	`deny_c_a1_overclaim, deny_safety_certification_overclaim, deny_legal_certification_overclaim`
MUT_IGNORE_RUNTIME_AUTHORITY_STATUS`	`B09b-09`	`hold_runtime_authority_missing`
MUT_IGNORE_SELF_EVO_WATCH_STATUS`	`B09b-10`	`hold_post_promotion_watch_missing`
MUT_IGNORE_CONTINUITY_UNKNOWN`	`B09b-11`	`hold_continuity_unknown, deny_continuity_rupture`
MUT_IGNORE_PUBLIC_CITATION_DRIFT`	`B09b-12`	`hold_public_citation_drift, hold_retracted_public_artifact_dependency`
MUT_IGNORE_INTEROP_CONFLICT`	`B09b-13`	`hold_interop_conflict_open`
MUT_ALLOW_CHECKER_PASS_AS_SAFETY`	`B09b-14`	`deny_checker_pass_as_safety, deny_fixture_pass_as_production_ready`
MUT_ALLOW_REPRODUCTION_AS_CERTIFICATION`	`B09b-15`	`deny_reproduction_pass_as_certification`
MUT_ALLOW_STANDARDS_MAPPING_AS_COMPLIANCE`	`B09b-16`	`deny_standards_mapping_as_compliance`
MUT_ALLOW_REGULATED_SUBMISSION_AS_APPROVAL`	`B09b-17`	`deny_regulated_submission_as_approval`
MUT_ALLOW_PUBLICATION_AS_DEPLOYMENT`	`B09b-18`	`deny_publication_as_deployment, deny_doi_as_deployment`
MUT_IGNORE_ROLLBACK_REQUIREMENT`	`B09b-19`	`hold_rollback_missing_for_production`
MUT_IGNORE_WATCH_REQUIREMENT`	`B09b-20`	`hold_watch_missing_for_field_deployment`
MUT_ALLOW_NEGATIVE_CACHE_HIT`	`B09b-21`	`quarantine_negative_cache_hit_allowed`
MUT_ALLOW_RED_PATTERN`	`B09b-22`	`quarantine_red_pattern_allowed`
MUT_ALLOW_PLUS_REPLACEMENT`	`B09b-23`	`deny_plus_replaced_by_deployment_record, deny_deployment_approval_as_binding_plus`
MUT_ALLOW_INSTITUTIONAL_INTEREST_AS_ENDORSEMENT`	`B09b-24`	`hold_institutional_interest_as_non_authority`
MUT_IGNORE_NEXT_RELEASE_DIRTY_CARRYOVER`	`B09b-25`	`hold_dirty_next_release_carryover`

The mutation matrix is deliberately one-fault oriented. Each mutation should have at least one one-fault fixture where the disabled guard is the reason the case is rejected, held, quarantined, or denied.

8. Decision expectations

The fixture pack uses this decision vocabulary:

```

ALLOW_CORPUS_ONLY
ALLOW_INTERNAL_TRIAL
ALLOW_OWNER_FIELD_PILOT
ALLOW_LIMITED_PRODUCTION
HOLD_FOR_EVIDENCE
REQUIRE_ANCHOR_ATTESTATION
REQUIRE_REGULATED_REVIEW
QUARANTINE_RELEASE
DENY_RELEASE

```

Expected semantics:

ALLOW_CORPUS_ONLY	artifact/corpus publication only; no runtime authority
ALLOW_INTERNAL_TRIAL	bounded internal/offline/sandbox trial
ALLOW_OWNER_FIELD_PILOT	owner/anchor-scoped field pilot, watched and revocable
ALLOW_LIMITED_PRODUCTION	limited production only with authority, rollback, and watch
HOLD_FOR_EVIDENCE	missing or unknown required evidence
REQUIRE_ANCHOR_ATTESTATION	approver role requires non-string attestation
REQUIRE_REGULATED_REVIEW	regulated review route required; not approval
QUARANTINE_RELEASE	red/negative-cache condition blocks release path
DENY_RELEASE	overclaim, prohibited collapse, or invalid authority path

9. String-role / attestation boundary

The fixture pack specifically attacks this failure class:

```

approver_role = HUMAN_ANCHOR
reviewer_role = REGULATED_REVIEWER
resolver_role = GOVERNANCE_QUORUM

```

when the role is only a string field.

Expected result:

```
APPROVER_ATTESTATION_MISSING -&gt; REQUIRE_ANCHOR_ATTESTATION
```

Valid deployment-layer approval requires a future ApproverAttestationRecord with a non-emulable source route. A fixture may not treat a declared role string as equivalent to anchor review, owner approval, regulated review, or governance quorum.

10. Red-pattern focus

The fixture pack treats the following as red or hold-worthy deployment patterns:

```
PUBLICATION_AS_DEPLOYMENT
CHECKER_PASS_AS_SAFETY
FIXTURE_PASS_AS_PRODUCTION_READY
INTEROP_PASS_AS_AUTHORITY
REPRODUCTION_PASS_AS_CERTIFICATION
STANDARDS_MAPPING_AS_COMPLIANCE
REGULATED_SUBMISSION_AS_APPROVAL
INSTITUTIONAL_INTEREST_AS_ENDORSEMENT
APPROVER_ROLE_STRING_ONLY
MODEL_APPROVER_LAUNDERING
TOOL_APPROVER_LAUNDERING
C_A1_DEPLOYMENT_OVERCLAIM
RUNTIME_AUTHORITY_UNKNOWN_ALLOWED
ROLLBACK_MISSING_FOR_PRODUCTION
WATCH_MISSING_FOR_FIELD_DEPLOYMENT
NEGATIVE_CACHE_HIT_ALLOWED
RED_PATTERN_ALLOWED
PLUS_REPLACED_BY_DEPLOYMENT_RECORD
NEXT_RELEASE_DIRTY_CARRYOVER
```

11. Acceptance commands

From the package root:

```
python3 scripts/run_09b_fixtures.py
python3 scripts/run_09b_mutations.py
sha256sum -c SHA256SUMS.txt
```

Expected summary for this seed package:

```
fixtures: 40/40 PASS
mutations: 25/25 CAUGHT
SHA256SUMS: OK
```

The fixture runner checks expectation consistency. It does not certify deployment.

12. Relationship to 09a

09a defines the machine-checkable deployment packet.

09b defines the fixture/mutation pressure surface that a future 09a checker must survive.

A future full executable package may merge 09a schemas with 09b fixture coverage. Until then, 09b is the hardening seed.

13. Next layer

The next natural deployment layer is:

```
09c_REGULATED_RELEASE_LEDGER_AND_WITHDRAWAL_DRILL_v0_1
```

Expected scope:

```
deployment candidate -&gt; review -&gt; bounded approval -&gt; release ledger -&gt; watch -&gt; incident / hold /
withdrawal / supersession -&gt; next-release gate
```

14. Public-safe statement

```
09b provides an adversarial fixture pack and mutation matrix for the c-calculus deployment profile.
It tests that publication, fixture success, checker success, reproduction success, standards mapping,
```

regulated submission, institutional interest, string-only approver roles, model/tool approval, negative-cache bypass, red-pattern bypass, dirty next-release carryover, and deployment paperwork cannot be laundered into deployment authority, safety certification, regulated approval, C-A1, or replacement of the governed binding operator +.
It is not legal advice, safety certification, deployment authorization, regulatory approval, standards compliance certification, C-A1 ratification, live substrate truth, or proof of completeness.

15. Closure formula

09a deployment packet
-> 09b fixture catalog
-> mutation matrix
-> guard coverage
-> future executable hardening

09b closes the first hardening pass for the deployment gate.

It keeps the central rule intact:

deployment may be tested;
it may not become +.

End of document.